

COMPX304-23A

Advanced Networking and Cybersecurity

Week 10
Symmetric key Distribution + Hardware Vulnerabilities

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Symmetric Key Encryption

- Provides confidentiality
- Is faster compared to alternatives
- Is suitable for fast hardware implementation
- But
 - How do we exchange secret keys



Symmetric Key Distribution

- A can select a key and physically deliver it to B.
- A third party can select the key and physically deliver it to A and B.
 - Usually called a Key Distribution Center
- If A and B have previously and recently used a key, one party can transmit the new key to the other, encrypted using the old key.
- If A and B each has an encrypted connection to a third party C, C can deliver a key on the encrypted links to A and B.



Needham-Schroeder Protocol

- Third-party key distribution protocol
- For session between A B mediated by KDC protocol overview is:
 1. $A \rightarrow KDC: ID_A || ID_B || N_1$
 2. $KDC \rightarrow A: E_{K_a}[Ks || ID_B || N_1 || E_{K_b}[Ks || ID_A]]$
 3. $A \rightarrow B: E_{K_b}[Ks || ID_A]$



Needham-Schroeder Protocol

- used to securely distribute a new session key for communications between A & B
- but is vulnerable to a replay attack if an old session key has been compromised
 - then message 3 can be resent convincing B that is communicating with A
- modifications to address this require:
 - timestamps
 - using an extra nonce



Key Distribution Solutions

- Key distribution solutions based on symmetric key requires a KDC
- They also require the communicating parties to trust the KDC.
- There are key distribution solutions that use asymmetric cryptography that don't require KDCs and trust in KDCs but require public key infrastructure (PKI)
- The alternative (although expensive) is key exchange based on a quantum phenomenon
- Algorithms based on quantum computing can help perform key exchange without the use of KDCs and PKI.



Meltdown and Spectre

Intel's processors have a security bug and the fix could slow down PCs

By Tom Warren | @tomwarren | Jan 3, 2018, 8:45am EST

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<https://www.theverge.com/2018/1/3/16844630/intel-processor-security-flaw-bug-kernel-windows-linux>



SPECTRE



MELTDOWN

116

ANDY GREENBERG SECURITY 01.03.2018 03:00 PM

A Critical Intel Flaw Breaks Basic Security for Most Computers

A Google-led team of researchers has found a critical chip flaw that developers are scrambling to patch in millions of computers.

<https://www.wired.com/story/critical-intel-flaw-breaks-basic-security-for-most-computers/>

THIS IS HUGE. REALLY. —

Meltdown and Spectre: Here's what Intel, Apple, Microsoft, others are doing about it

Intel, Microsoft, ARM, and others have responded. We dig in.

PETER BRIGHT - 1/6/2018, 2:52 AM

<https://arstechnica.com/gadgets/2018/01/meltdown-and-spectre-heres-what-intel-apple-microsoft-others-are-doing-about-it/>

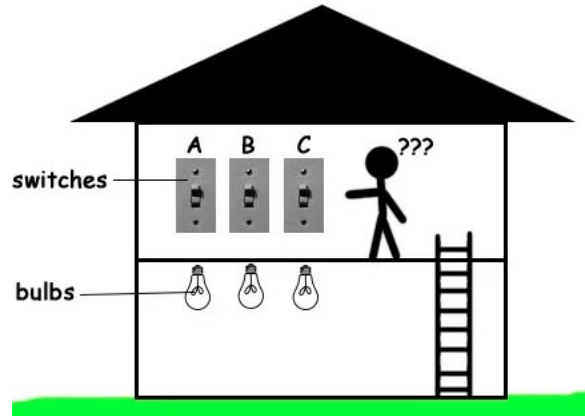


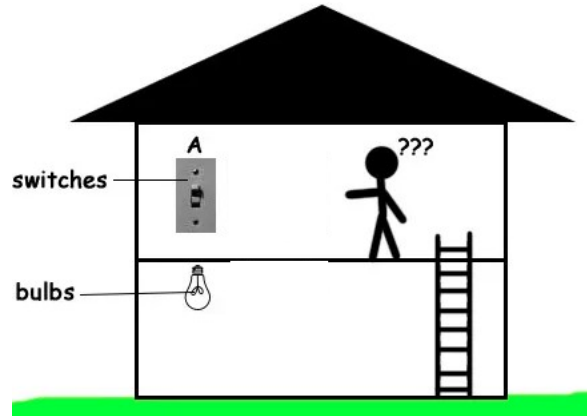
CROW Lab Cybersecurity Researchers of Waikato

Meltdown and Spectre

- Confidentiality attacks
- Extracts private information
 - Bit by bit
- Spectre
 - Breaks isolation
- Meltdown
 - Breaks isolation between user applications and the OS.







For a technical understanding

- we need to know about:
 - Computer Architecture
 - Computer Organization



Computer Architecture

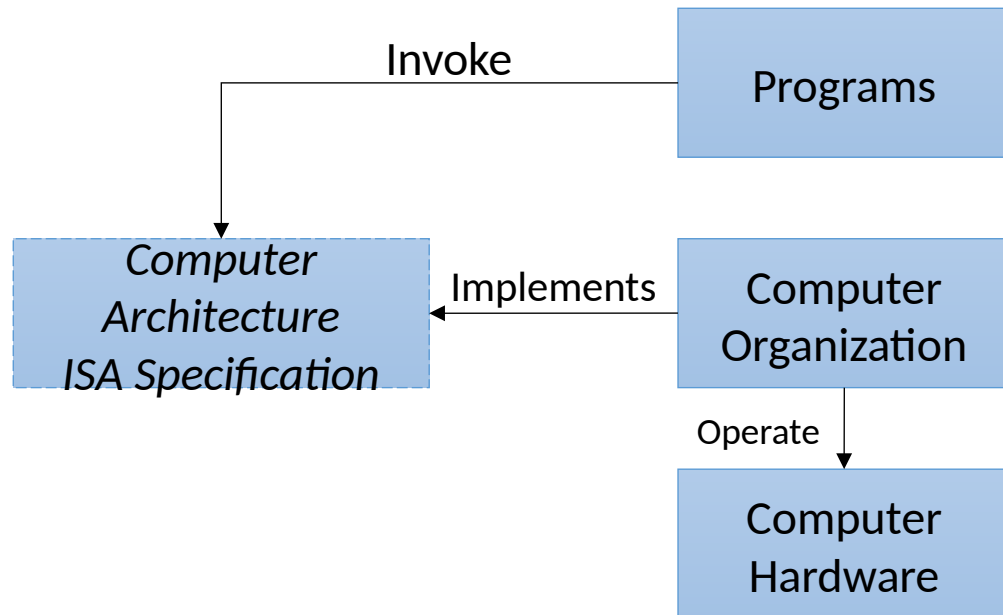
- The interface the CPU implements
- Also known as Instruction Set Architecture (ISA)
 - Load, store, add, etc.
- Programs execute in it. The CPU always repeats:
 - Fetch the next instruction
 - Decode it
 - Execute it

```
01 .MODEL SMALL
02 .STACK 100H
03 .CODE
04
05 MOV AX, 0x3C
06 MOV BX, 0000000000001010B
07 ADD AX, BX
08 MOV BX, 14
09 SUB AX, BX
10
11 MOV AH, 04CH
12 INT 21H
```



Computer Organization

- The mechanics under the ISA
- Transparent to the user
- But physical/real
- Not just a spec



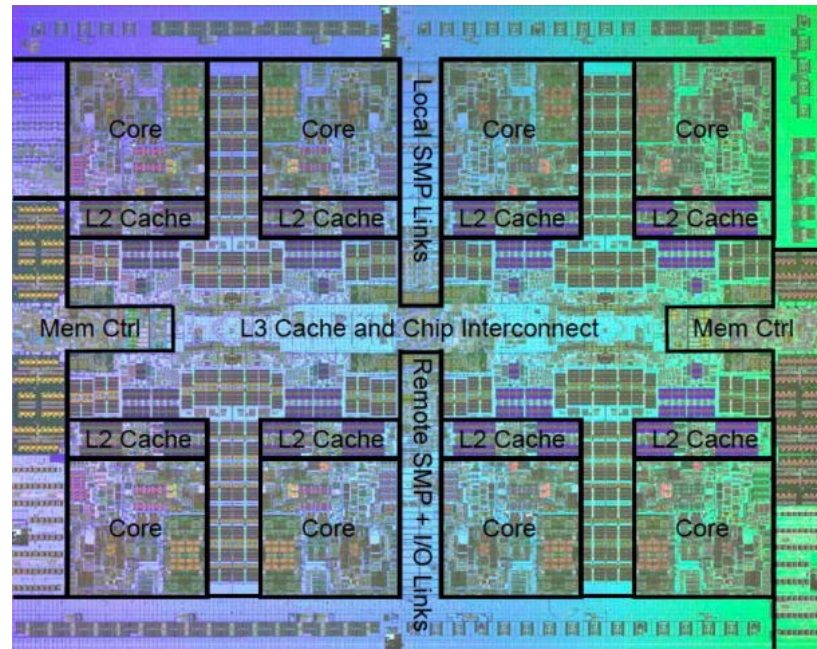
CPU Cache

- Main memory is much slower than the CPU
- We do not want the CPU to be idle for long periods of time
- Cache is a smaller but faster piece of memory closer to the CPU
- Every time something is fetched from the main memory it is also stored in the cache
- 80:20 rule



CPU Cache

- A load 0x... is not just a load!
 1. The address is converted to its physical address and permissions are checked
 2. The address is looked up in the Level-1 (L1) cache. If not there,
 3. It's looked up in the L2 cache. If not there,
 4. In the L3 cache, if not there
 5. From main memory (cache miss)
- Each of these pathways take a significantly different time (CPU cycles) to complete



<https://arstechnica.com/gadgets/2009/09/ibms-8-core-power7-twice-the-muscle-half-the-transistors/>

